

USING RESOURCES – CERAMICS, POLYMERS & COMPOSITES

KEY VOCABULARY

Ceramic: a hard, brittle material like clay pottery or glass.

Thermosoftening: polymer that melts and remoulds when heated.

Thermosetting: polymer with cross-links — does not melt.

Composite: a material made of two materials with different properties.

COMMON MISCONCEPTIONS

~~— All polymers melt when heated.~~

✓ **Thermosetting polymers do not melt.**

~~— Composites are a single material.~~

✓ **They combine a matrix and reinforcement.**

~~— Ceramics conduct electricity well.~~

✓ **Ceramics are insulators.**

MATCHING PROPERTIES

Ceramics: hard, brittle, insulators (pottery, glass).

Thermosoftening: melts to remould (e.g. poly(ethene)).

Thermosetting: rigid, cross-linked — won't melt.

Composite: matrix + reinforcement (e.g. fibreglass).

Choose the material to fit the job.

1

CERAMIC PROPERTIES

State two typical properties of a **ceramic**. [2 marks]

2

TWO POLYMERS

Explain the difference between thermosoftening and thermosetting polymers. [3 marks]

3

DEFINE COMPOSITE

State what is meant by a **composite** material. [1 mark]

4

CHOOSE A MATERIAL

Suggest a suitable material for a saucepan handle and why. [2 marks]

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Thermosetting polymers have _____ so they do not _____ when heated.

cross-links

melt

soften

ceramic

6

COMPOSITE EXAMPLE

Give one composite material and a use. [2 marks]